

Chapter 7: Bond Valuation and Yield Analysis - Practice Questions

1- A bond matures in 15 years, has a par value of \$1,000, and pays an annual coupon of 6%. If the market interest rate is 8%, what is the price of the bond?

2- A bond matures in 12 years, has a par value of \$1,000, and pays an annual coupon of \$90. If the market interest rate is 6%, what is the price of the bond?

3- A zero-coupon bond matures in 5 years with a par value of \$1,000. If the market rate of interest is 7%, what is the current price of the bond?

4- Acme Corporation's bonds are currently selling for \$850. The bonds have 10 years to maturity, an annual coupon rate of 7%, and a par value of \$1,000. What is their yield to maturity (YTM)?

5- Beta Corp's bonds are currently priced at \$1,100. They pay an annual coupon of \$80, have a 15-year maturity, and a par value of \$1,000. What is the yield to maturity (YTM)?

6- Omega Industries has bonds outstanding that pay a \$100 annual coupon, have 25 years until maturity, and a par value of \$1,000. The bonds can be called in 5 years at \$1,050, and they currently sell for \$1,200. What is the yield to call (YTC), assuming they are called at the earliest opportunity?

7- A bond from Delta Company pays an \$85 annual coupon, matures in 20 years, and has a call option in 8 years at \$1,050. The current market rate is 7%. What is the likely price of this bond today?

8- A bond matures in 18 years, pays a \$50 semiannual coupon, and has a par value of \$1,000. If the market interest rate is 9% compounded semiannually, what is the price of the bond?

9- Zeta Corporation has bonds currently selling for \$950 in the market. Each bond pays an annual coupon payment of \$75, based on a par value of \$1,000. Investors are currently requiring a 9% annual return on similar bonds. Assuming the bond makes coupon payments **once a year**, calculate the **number of years remaining to maturity (N)** for Zeta Corporation's bonds.

10- Theta Inc. issues bonds with \$40 semiannual coupon payments. The bonds have 16 years until maturity, a par value of \$1,000, and sell for \$950. What is the yield to maturity (YTM) on a **semiannual basis**?

11- A zero-coupon bond matures in 3 years with a par value of \$1,000. It's selling today for \$850. If the company's marginal tax rate is 35%, what is the annual after-tax cost of debt to the company?

12- Sigma Inc.'s bonds have a 10% annual coupon, 30 years to maturity, and can be called in 7 years at \$1,100. The bonds currently sell for \$1,250. The market rate is 7%.

a) What is the yield to maturity (YTM)?

b) What is the yield to call (YTC)?